

Practical Design of High-Density Bend Waveguide Array for Co-Packaged Fan-Out Routing

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Abstract An optimised bend array under limited NA constraints reduces the minimum radius of curvature by 21% at 0.3 dB/turn and achieves -36 dB crosstalk ($L/S = 2.5/40 \mu\text{m}$), enabling dense FO routing of 576 channels within a 15.0 mm length without major changes to existing package designs. ©2026 The Author(s)

Introduction

Co-packaged optics (CPO) are increasingly adopted to address the I/O bandwidth bottleneck in large-scale AI/ML systems [1,2]. However, the bandwidth density of optical engine (OE) is limited by the outer diameter of single-mode fibers (SMFs) in conventional fiber arrays (FAs) [3]. Equivalent density can be increased by stacking SMFs in multiple layers, but stacking beyond three layers is difficult because of fiber diameter variation. Using optical waveguides (OWGs) for FO routing can provide a breakthrough by enabling a narrower pitch [4,5]. For example, to reach 10 Tbps/mm at 448 Gbps per fiber, stacking requires three layers, whereas FO routing can achieve this density at a pitch of around $40 \mu\text{m}$ as shown in Fig.1.

FO routing must still satisfy footprint, bend loss, and bend crosstalk constraints within the limited area of FC-BGA packages. Theoretically, increasing the numerical aperture (NA) to enhance mode confinement is a simple approach to this issue, but a smaller core typically makes single-mode operation difficult and may reduce manufacturability. In addition, a smaller mode field diameter (MFD) reduces coupling tolerance[6]. Therefore, the key challenge is to meet FO specifications under limited-NA conditions.

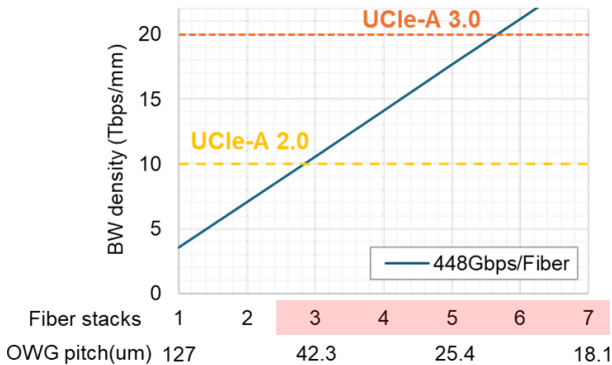


Fig. 1 Fiber stacking vs. FO routing with waveguides for high shoreline density

To address these design constraints, we propose and optimize the bend geometry which dominates both radiation loss and bend-induced crosstalk in narrow-pitch OWGs. Establishing a low-loss and low-crosstalk (XT) bend under limited-NA conditions provides the foundation for scalable FO routing. Secondly, we propose an FO-routing optimization framework that maintains minimum curvature and the required pitch while enabling routing within a practical footprint. A complete design example is provided to demonstrate feasibility for next-generation CPO packages. The following section describes our optimization framework.

Optimal design using Bayesian optimization

Bend analysis under limited NA requires a large simulation domain, which increases computing cost. Therefore, 3D full vectorial FDTD or FDFD is impractical for OWG bend structures, and inverse design based on an adjoint method is also difficult for the same reason. In this study, parametric optimization based on the eigenmode expansion (EME) method is adopted because EME is an efficient approximate solver based on the slowly varying envelope approximation (SVEA). For bend structures, conformal mapping is applied to reduce the simulation domain while maintaining accuracy. However, the cost remains high when the structure requires a fine division size due to varying widths along the propagation

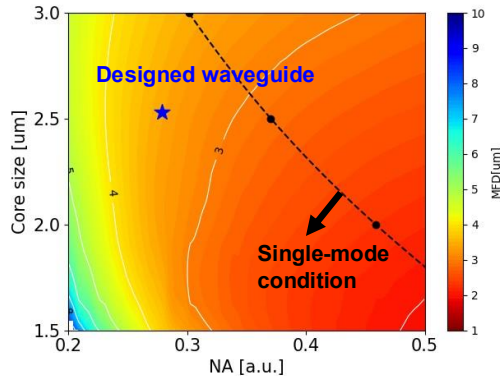


Fig. 2 OWG specifications on the design map

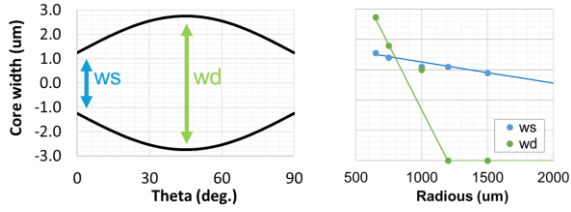


Fig. 3 Hyperparameter definition and optimized parameters for each radius of curvature

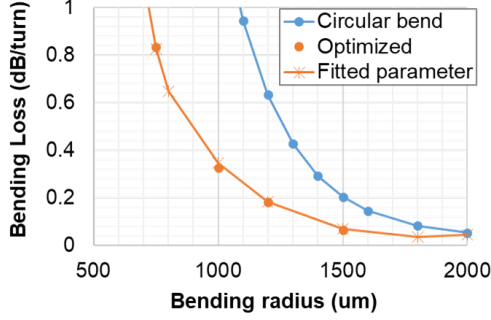


Fig. 4 Bend loss with fitted parameters

direction, which limits the number of optimization trials. To calculate efficiently, we adopt Bayesian optimization. The structure is described with a small number of variables. The core widths at the start and middle points, W_s and W_d respectively, are defined as design variables in Eq. (1). The objective function is the TE-mode transmission efficiency at 1310 nm, which yields a maximization problem.

$$W(z) = W_s + W_d \sin(\pi z) \mid z \in [0,1] \quad (1)$$

For each bend radius, Bayesian optimization was run for 60 iterations, and the extracted parameters are summarized in Fig.3. Compared with a circular bend, the optimized bend reduced the minimum radius by 21% at 0.3 dB/turn.

For FO routing, the bend geometry should vary smoothly across adjacent channels[7]. Therefore, a design rule is required for an arbitrary radius. To evaluate the general applicability of the designs, linear regression was applied to the parameters obtained at each radius, then the fitted parameters were evaluated by simulation. The fitted parameters performed similarly to the original optimized designs in Fig.4.

Bend crosstalk analysis of the optimized structure

To support a narrower channel pitch with tight bends, it is essential to evaluate crosstalk between optimized waveguides. Conventionally, supermode methods[7] or analytical approach using conformal mapping[8] are available for circular bends, but the optimized structure in this work has a varying structure along the propagation direction. Therefore, the phase mismatch accumulated along z must be considered explicitly. When the propagation constant depends on z ,

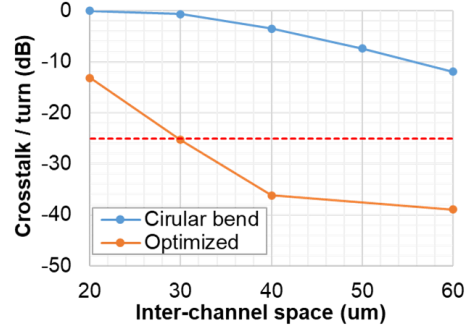


Fig. 5 Bend crosstalk for each structure

the accumulated phase difference at an arbitrary position is given by Eq. (2).

$$\phi(z) = \int_0^z \beta_p(z') - \beta_q(z') dz' \quad (2)$$

Using the phase term above, the crosstalk can be written as Eq. (3).

$$\frac{\partial E_p}{\partial z} = -j\kappa_{pq}(z)E_q \exp(-j\phi(z)) \quad (3)$$

coupling coefficient is given by

$$\kappa_{pq} = \frac{1}{4P_p} \iint \Delta n^2 E_p^*(x, y; z) E_q(x, y; z) dx dy \quad (4)$$

where $\Delta n_{clad}^2 = n(x, y; z)^2 - n_{clad}^2$. In tight bends, the guided mode becomes asymmetric, and direct coupled-mode evaluation may be numerically unstable due to non-Hermitian properties. Therefore, we use $\kappa = \kappa_{12}$ approximation for conservative evaluation. This is because in tight bends the coupling is predominantly directed from the inner channel to the outer channel, especially over a distance shorter than the coupling length.

Complex field profiles and propagation constants are obtained from a mode solver, the coupling coefficient is calculated from overlap integrals, and the z -dependent coupled-mode equations are solved by Runge–Kutta integration.

Two cases with similar bend loss were compared. A circular bend with $R = 1500 \mu\text{m}$ and an optimized bend with $R = 1000 \mu\text{m}$. The optimized structure consistently exhibits lower crosstalk across channel spacings, achieving -36 dB at an L/S configuration of $2.5/40 \mu\text{m}$, as shown in Fig.5.

The improvement is attributed to two factors. Increased mode confinement and a non-uniform structure along the propagation direction. The non-uniform structure causes both the complex coupling coefficient and the propagation constant difference $\Delta\beta$ to vary with z , which can introduce phase mismatch and partial cancellation of coupled mode, thereby reducing the coupling efficiency. As shown in Fig.6, the real part of the coupling coefficient changes from negative to positive, and $\Delta\beta$ becomes large near the middle of the structure. These results indicate that the optimization reduces both bend loss and bend crosstalk.

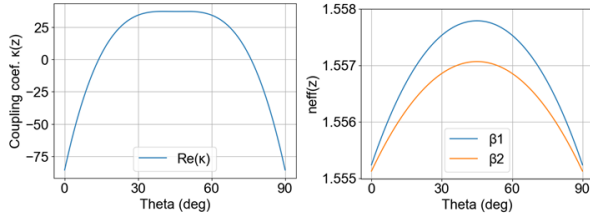


Fig. 6 Coupling coefficient and propagation constant difference ($\Delta\beta$) along a 90° turn

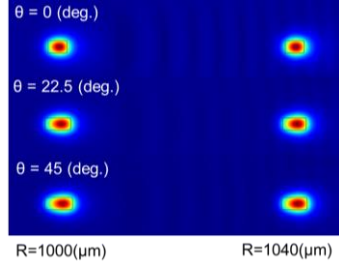


Fig. 7 Distribution of $\text{Re}(E(z))$

These results demonstrate that the proposed bend model enables high density FO routing. We next discuss practical design feasibility and whether the available FO length is sufficient for an FC-BGA package.

FO routing optimization and design example

Fan-out (FO) routing with a large number of channels is generally a NP-hard layout problem because the routing must satisfy pitch, curvature, and footprint constraints simultaneously.

To make the design practical, we introduce a set of geometric constraints that ensure manufacturability while enabling compact routing. Specifically, the minimum pitch between adjacent channels is preserved, and a continuous curvature condition is applied so that neighbouring channels can be routed without violating the pitch requirement.

The curvature difference between adjacent channels, ΔR , is set to p_1 which is the I/O pitch on the PIC side. The curvature radius of the n -th channel is then given by Eq. (5). Based on this rule, we derive a minimum-FO condition under a bend angle constraint and identify the bottleneck channel that determines the required routing length.

$$R_n = R_{min} + p_1(n - 1) \quad (5)$$

To define the minimum FO routing length, we impose a bend-angle constraint of $\theta \leq 90^\circ$. The bottleneck occurs when $L = 2R_i$. To design a crossing angle of 90° , the required pitch-change condition becomes $\Delta_n \geq 2R_n$. Substituting into Eq. (5) yields Eq. (6), which determines the critical radius R_i and thus the minimum routing length.

$$\Delta_n \geq 2\{R_{min} + p_1(n - 1)\} \quad (6)$$

where Δ_n is the gap between p_1 and p_2 in an arbitrary channel. From Eq. (6), R_c is determined by the smallest channel index that satisfies the

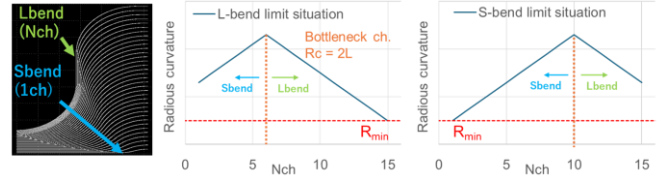


Fig. 8 Basic approach for FO design automation

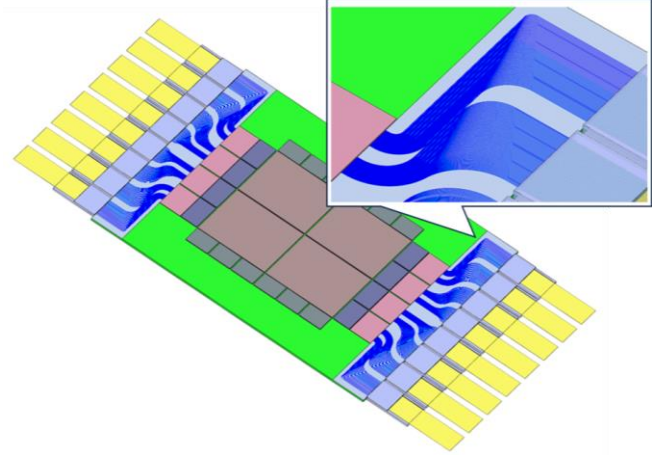


Fig. 9 The schematic of CPO package with optimized OWG routing

condition and by the total number of channels. The minimum FO routing length is then determined. In practice, the limitation is first identified as either an S-bend or L-bend. The design can then start from the outer channel and gradually relax the curvature to satisfy the pitch condition.

As a design example, the method was applied to FO routing from $144 \text{ channels} \times 4 \text{ OEs}$ with a $40 \mu\text{m}$ optical I/O pitch to $72\text{-channel} \times 8 \text{ FAs}$ with $127 \mu\text{m}$ pitch. Under the optimized condition, the FO routing length was 15.0 mm , and the package layout is shown in Fig.9. This result indicates that FO routing can be implemented without major changes to the existing package design.

Conclusion

This paper presents a Bayesian-optimization-based parametric approach that simultaneously reduces bend loss and bend crosstalk under limited-NA conditions while remaining compatible with circular-bend layouts.

We also proposed a deterministic FO-routing method that satisfies pitch and curvature constraints. In a practical CPO layout, high-density OWG routing for 576 channels was achieved within a 15.0-mm routing length inside an FC-BGA footprint, confirming practicality under packaging constraints.

Together, limited-NA bend optimization and deterministic FO routing provide a scalable path to higher shoreline bandwidth density without major changes to existing package architectures.

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